Linear Acoustics with Advection and Multirate Time Integration

Split-explicit, exponential, Lawson, averaging and resonance-based methods

Your Name

Your Institute

October 12, 2026

Outline

1. Motivation: time integration in NWP
2. Model problem
3. Split-explicit methods as ETD
4. Lawson, averaging and resonance methods
5. Application: staggered grids, high-order advection
6. Summary

Why this problem? Time stepping in numerical weather models

- Compressible (non-hydrostatic) dynamical cores solve the Euler equations on the sphere/in a box.
- Wide range of time scales in one system:

process	speed	role
sound waves	$c \approx 340\,\mathrm{m/s}$	fast, *not* meteorologically relevant
gravity waves	~ 30–$100\,\mathrm{m/s}$	fast/intermediate
advection by wind	$U \approx 10$–$50\,\mathrm{m/s}$	slow, *relevant*

- Fully explicit methods: step size limited by $c\,\Delta t/\Delta x \lesssim 1 \;\Rightarrow$ far smaller than accuracy needs for the slow, relevant dynamics.
- Fully implicit: global (Helmholtz-type) solves, poor scalability on massively parallel machines.

Key idea

Treat the fast part cheaply/exactly and the slow part with a large step: **multirate** / **split** / **exponential** methods.

Where it is used and what is needed

- Split-explicit time stepping (Klemp–Wilhelmson; Wicker–Skamarock RK3 with acoustic substeps) is the standard in WRF, COSMO/ICON-type and many other cores.
- Exponential integrators and multirate schemes are active research directions for dynamical cores.
- The **linear acoustic–advection system** is the core subproblem:
  - it contains exactly the scale separation $c \gg U$ (Mach number $M = U/c \approx 0.1$),
  - it allows analysis in Fourier space (stability, dispersion, resonance),
  - errors of any scheme for the full nonlinear system must at least be good here.

Plan of the talk

1. model problem and the abstract splitting $\mathrm{q}' = \mathcal{L}\mathrm{q} + \mathcal{N}\mathrm{q}$
2. split-explicit methods as a special case of ETD
3. Lawson, averaging and resonance-based methods
4. application: staggered grids in 1D and 2D with high-order advection

Linear acoustics with advection

Perturbations (p, u) about a constant background state (ρ_0, U), speed of sound c:

$$\partial_t u + U\,\partial_x u + \tfrac{1}{\rho_0}\,\partial_x p = 0,$$
$$\partial_t p + U\,\partial_x p + \rho_0 c^2\,\partial_x u = 0.$$

Write $\mathrm{q} = (p, u)^\top$ and split

$$\partial_t \mathrm{q} = \underbrace{-\begin{pmatrix} 0 & \rho_0 c^2 \\ 1/\rho_0 & 0 \end{pmatrix}\partial_x \mathrm{q}}_{\mathcal{L}\mathrm{q}:\ \text{acoustic (fast)}} \underbrace{-\,U\,\partial_x \mathrm{q}}_{\mathcal{N}\mathrm{q}:\ \text{advection (slow)}}.$$

- $\mathcal{L}$: eigenvalues in Fourier space $\pm \mathrm{i}ck$ (purely oscillatory, large).
- $\mathcal{N}$: eigenvalue $-\mathrm{i}Uk$ (small, $|U| \ll c$).
- For constant U on a periodic domain the two operators **commute**, also after discretization with the same stencil for p and u (all operators are circulant). Non-commutativity enters via variable $U(x)$, boundaries, or flux-form/interpolated discretizations.

Abstract setting

$$\mathrm{q}'(t) = \mathcal{L}\mathrm{q}(t) + \mathcal{N}\mathrm{q}(t), \qquad \mathrm{q}(t_0) = \mathrm{q}_0,$$

after spatial discretization: $\mathcal{L}, \mathcal{N}$ are matrices, $\|h\mathcal{L}\| \gg \|h\mathcal{N}\|$ for the time steps of interest.

Variation of constants over one step $[t_n, t_n + h]$:

$$\mathrm{q}(t_n + h) = \mathrm{e}^{h\mathcal{L}}\mathrm{q}_n + \int_0^h \mathrm{e}^{(h-s)\mathcal{L}}\,\mathcal{N}\,\mathrm{q}(t_n + s)\,\mathrm{d}s.$$

- The fast part is treated *exactly* (or very accurately) through $\mathrm{e}^{h\mathcal{L}}$.
- Only the slow part $\mathcal{N}\mathrm{q}$ must be approximated; its time scale is $\sim 1/(Uk)$.
- Different multirate methods = different ways to approximate this integral.

Notation: $\varphi_1(z) = \dfrac{\mathrm{e}^z - 1}{z}, \quad \varphi_2(z) = \dfrac{\varphi_1(z) - 1}{z}, \quad \varphi_{k+1}(z) = \dfrac{\varphi_k(z) - 1/k!}{z}.$

Exponential time differencing (ETD)

Approximate $\mathcal{N}\mathsf{q}(t_n + s)$ by a polynomial in s and integrate the exponential exactly.

ETD1 (exponential Euler):

$$\mathsf{q}_{n+1} = \mathrm{e}^{h\mathcal{L}}\mathsf{q}_n + h\,\varphi_1(h\mathcal{L})\mathcal{N}\mathsf{q}_n.$$

ETD2 / ETDRK2 (Cox–Matthews type):

$$\begin{aligned}\mathsf{a}_n &= \mathrm{e}^{h\mathcal{L}}\mathsf{q}_n + h\,\varphi_1(h\mathcal{L})\mathcal{N}\mathsf{q}_n,\\ \mathsf{q}_{n+1} &= \mathsf{a}_n + h\,\varphi_2(h\mathcal{L})\big(\mathcal{N}\mathsf{a}_n - \mathcal{N}\mathsf{q}_n\big).\end{aligned}$$

- Exact for $\mathsf{q}' = \mathcal{L}\mathsf{q} + \mathsf{g}$ with constant g ("frozen slow tendency").
- Requires $\mathrm{e}^{h\mathcal{L}}\mathsf{v}$ and $\varphi_k(h\mathcal{L})\mathsf{v}$: Krylov / Fourier / (rational) approximations.
- Acoustic oscillations: no CFL restriction from c at all (but see resonances later).

Split-explicit time stepping

Klemp–Wilhelmson / Wicker–Skamarock: freeze the slow tendency during the step and integrate the fast system by m small substeps $\tau = h/m$ (forward–backward for the acoustic part):

$$z_0 = q_n, \qquad z_{j+1} = z_j + \tau\big(\mathcal{L} z_j + g\big), \quad g = \mathcal{N} q_n, \qquad q_{n+1} = z_m.$$

Since $z' = \mathcal{L} z + g$ has the exact solution $e^{t\mathcal{L}} q_n + t\,\varphi_1(t\mathcal{L}) g$,

$$\begin{aligned} z_m &= (I + \tau\mathcal{L})^m q_n + \tau \sum_{j=0}^{m-1} (I + \tau\mathcal{L})^j\, g \\ &\approx e^{h\mathcal{L}} q_n + h\,\varphi_1(h\mathcal{L}) g. \end{aligned}$$

Split-explicit = ETD1 with substepped approximations

$e^{h\mathcal{L}} \;\to\; R_m(\tau\mathcal{L}) = (I + \tau\mathcal{L})^m$ and $\varphi_1(h\mathcal{L}) \;\to\; \frac{1}{m}\sum_{j<m}(I + \tau\mathcal{L})^j$. The same view applies to the RK3 outer loop of Wicker–Skamarock (stages with $h/3$, $h/2$, h).

Consequences of the ETD view

- **Order:** the slow forcing is frozen $\Rightarrow$ first order for the pure ETD1 version; higher order through RK-type outer stages or ETDRK schemes with φ_2, φ_3.
- **Fast part inexact:** $\mathrm{e}^{h\mathcal{L}}$ is replaced by a stability-limited polynomial; the substep count m controls both accuracy and stability (acoustic Courant number $c\tau/\Delta x \leq C$).
- **Free choice of "fast solver"**: substepping, Fourier/FFT (periodic), Krylov, Chebyshev – all realize the same ETD framework.
- **Analysis in Fourier space:** on each mode, $\mathcal{L} \to \pm \mathrm{i}ck$, $\mathcal{N} \to -\mathrm{i}Uk$ and all $\varphi_k(\mathrm{i}\theta)$ with $\theta = \pm ckh$ are explicit; amplification factors can be compared with $\mathrm{e}^{-\mathrm{i}(\pm c - U)\, kh}$.

Lawson (integrating factor) methods

Change variables $\mathsf{v}(t) = \mathrm{e}^{-(t-t_n)\mathcal{L}}\mathsf{q}(t)$:

$$\mathsf{v}'(t) = \mathrm{e}^{-(t-t_n)\mathcal{L}}\,\mathcal{N}\,\mathrm{e}^{(t-t_n)\mathcal{L}}\mathsf{v}(t) =: \widetilde{\mathcal{N}}(t)\,\mathsf{v}(t),$$

apply an explicit Runge–Kutta method to v and transform back.

Lawson–Euler:

$$\mathsf{q}_{n+1} = \mathrm{e}^{h\mathcal{L}}\big(\mathsf{q}_n + h\mathcal{N}\mathsf{q}_n\big).$$

Lawson–RK4: classical RK4 with $\mathrm{e}^{c_i h\mathcal{L}}$ weights, e.g. stage values $\mathsf{Y}_i = \mathrm{e}^{c_i h\mathcal{L}}\mathsf{q}_n + h\sum_j a_{ij}\,\mathrm{e}^{(c_i-c_j)h\mathcal{L}}\mathcal{N}\mathsf{Y}_j$.

- Simple to implement: only matrix exponentials of $\mathcal{L}$ (diagonal in Fourier space).
- Order conditions are those of the underlying RK method (for sufficiently smooth solutions).
- Drawbacks: does not preserve steady states of $\mathcal{L}+\mathcal{N}$; error constants depend on $\|\mathcal{N}\|$, not on a well-adapted φ-function structure; stiff order reduction possible.

Efficient Lawson-RK4

Classical RK4 on $\mathsf{v}' = \widetilde{\mathcal{N}}(t)\mathsf{v}$ with $E = \mathrm{e}^{h\mathcal{L}/2}$ gives

$$\mathsf{q}_{n+1} = E^2\mathsf{q}_n + \tfrac{h}{6}\big(E^2\mathsf{k}_1 + 2E(\mathsf{k}_2 + \mathsf{k}_3) + \mathsf{k}_4\big),$$

with stages $\mathsf{Y}_1 = \mathsf{q}_n$, $\mathsf{Y}_2 = E(\mathsf{q}_n + \tfrac{h}{2}\mathsf{k}_1)$, $\mathsf{Y}_3 = E\mathsf{q}_n + \tfrac{h}{2}\mathsf{k}_2$, $\mathsf{Y}_4 = E^2\mathsf{q}_n + hE\mathsf{k}_3$, $\quad \mathsf{k}_i = \mathcal{N}\mathsf{Y}_i$.

Reorganized with E applied to sums (using linearity of E):

$$\begin{aligned}
&\mathsf{k}_1 = \mathcal{N}\mathsf{q}_n, && \mathsf{w} = E\mathsf{q}_n, && \mathsf{Y}_2 = E\big(\mathsf{q}_n + \tfrac{h}{2}\mathsf{k}_1\big), && \mathsf{k}_2 = \mathcal{N}\mathsf{Y}_2,\\
&\mathsf{Y}_3 = \mathsf{w} + \tfrac{h}{2}\mathsf{k}_2, && \mathsf{k}_3 = \mathcal{N}\mathsf{Y}_3, && \mathsf{Y}_4 = E\big(\mathsf{w} + h\mathsf{k}_3\big), && \mathsf{k}_4 = \mathcal{N}\mathsf{Y}_4,
\end{aligned}$$

$$\mathsf{q}_{n+1} = E\Big[\mathsf{w} + \tfrac{1}{3}\big(\mathsf{Y}_2 - \mathsf{w}\big) + \tfrac{h}{3}\big(\mathsf{k}_2 + \mathsf{k}_3\big)\Big] + \tfrac{h}{6}\mathsf{k}_4.$$

- **4 evaluations of $\mathcal{N}$ and 4 applications of E** per step; $\mathrm{e}^{h\mathcal{L}} = E^2$ is never formed ($\tfrac{1}{3}(\mathsf{Y}_2 - \mathsf{w}) = \tfrac{h}{6}E\mathsf{k}_1$ reuses Y_2).
- E is precomputed once: on the staggered grid a 2×2 block per wavenumber, applied by FFT.

Averaging methods

Observation: in the interaction picture the slow forcing is

$$\widetilde{\mathcal{N}}(s) = \mathrm{e}^{-s\mathcal{L}}\mathcal{N}\,\mathrm{e}^{s\mathcal{L}},$$

which oscillates with the fast (acoustic) frequencies $\pm ck$. Its time average

$$\overline{\mathcal{N}}_h = \frac{1}{h}\int_0^h \mathrm{e}^{-s\mathcal{L}}\mathcal{N}\,\mathrm{e}^{s\mathcal{L}}\,\mathrm{d}s$$

carries the net slow effect; oscillating contributions average out.

Averaged integrator (first order sketch):

$$\mathsf{q}_{n+1} = \mathrm{e}^{h\mathcal{L}}\Big(I + h\overline{\mathcal{N}}_h\Big)\mathsf{q}_n \quad \text{or} \quad \mathsf{q}_{n+1} = \mathrm{e}^{h(\mathcal{L}+\overline{\mathcal{N}}_h)}\mathsf{q}_n.$$

- For commuting $\mathcal{L}, \mathcal{N}$: $\overline{\mathcal{N}}_h = \mathcal{N}$, and the scheme is exact in the continuum.
- For non-commuting operators (variable wind, interpolated fluxes, boundaries), $\overline{\mathcal{N}}_h$ captures the leading commutator corrections $[\mathcal{L}, \mathcal{N}]$.
- The average can be computed by quadrature in s, by a Fourier/eigenbasis formula, or by substeps.

Resonances

On a Fourier mode with $\mathcal{L} \to \pm \mathrm{i}\omega$, $\omega = ck$, the exponential integrator terms contain

$$\varphi_1(\mathrm{i}\theta) = \frac{\mathrm{e}^{\mathrm{i}\theta} - 1}{\mathrm{i}\theta}, \qquad \theta = \omega h,$$

and error constants contain factors like $1/\sin(\theta/2)$ in multistep/mean-value type schemes.

Step-size resonances

When $\omega h \approx 2\pi n$ (the step is a multiple of the acoustic period), errors can be amplified (in the worst case, instability) for some exponential/Lawson schemes.

Resonance-based (filtered) methods:

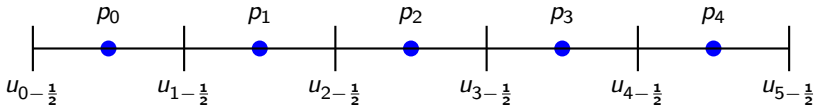
- expand the Duhamel integral and integrate the oscillatory factors $\mathrm{e}^{\mathrm{i}s(\pm\omega_1 \pm \omega_2)}$ *exactly*;
- resonant combinations ($\omega_1 \pm \omega_2 \approx 0$) are kept as they are, non-resonant ones are integrated analytically;
- result: error bounds that do not depend on ωh and no step-size resonance.

Comparison of the schemes (what we will test)

	fast part	slow part	resonance
split-explicit	substepped polynomial	frozen / RK stages	none (stability limit)
ETD(RK)	exact $\mathrm{e}^{h\mathcal{L}}$, φ_k	polynomial in s	possible
Lawson	exact $\mathrm{e}^{h\mathcal{L}}$	RK on interaction picture	possible
averaging	exact $\mathrm{e}^{h\mathcal{L}}$	time-averaged $\overline{\mathcal{N}}_h$	reduced
resonance-based	exact $\mathrm{e}^{h\mathcal{L}}$	oscillatory integrals exact	avoided

- Criteria: order in h, accuracy vs. ωh (large acoustic steps), cost (number of $\mathcal{L}$-evaluations / FFTs / substeps), robustness near $ckh = 2\pi n$.

Spatial discretization: staggered grid (1D)



- p at cell centers, u at cell faces (Arakawa C in 2D): compact, accurate acoustic operator

$$(\partial_x u)_j \approx \frac{u_{j+1/2} - u_{j-1/2}}{\Delta x}, \qquad (\partial_x p)_{j+1/2} \approx \frac{p_{j+1} - p_j}{\Delta x} \quad \text{(2nd order)}.$$

- Higher order: wider centered staggered stencils (4th/6th order) for $\mathcal{L}$.
- **Advection** $\mathcal{N} = -U\partial_x$: both fields must be advected on their own grid points $\Rightarrow$ high-order upwind-biased stencils (e.g. 3rd/5th order). With constant U and the same stencil on both arrays, $\mathcal{L}$ and $\mathcal{N}$ are circulant and commute: the exact semi-discrete solution is known (used as reference below).

High-order advection operators

Upwind-biased reconstruction of face values $\hat{\phi}_{j+1/2}$ (for $U > 0$), advective operator

$$(\mathcal{N}\phi)_j = -\frac{U}{\Delta x}\left(\hat{\phi}_{j+1/2} - \hat{\phi}_{j-1/2}\right).$$

order	face reconstruction
1st	$\hat{\phi}_{j+1/2} = \phi_j$
3rd	$\hat{\phi}_{j+1/2} = \frac{1}{6}\left(-\phi_{j-1} + 5\phi_j + 2\phi_{j+1}\right)$
5th	$\hat{\phi}_{j+1/2} = \frac{1}{60}\left(2\phi_{j-2} - 13\phi_{j-1} + 47\phi_j + 27\phi_{j+1} - 3\phi_{j+2}\right)$

- Odd orders: built-in dissipation (upwinding) $\Rightarrow$ implicit damping of grid-scale noise.
- Time integrator error must stay below the spatial error: low-order splitting would spoil the high-order advection.
- Fourier symbol of the discrete operator: $\mathcal{N} \to -\mathrm{i}U\,\kappa(k\Delta x)/\Delta x$, used for analysis of all schemes above.

Test setup: 1D

- Periodic domain $[0, L]$, background U, c with $M = U/c \in \{0.05, 0.1, 0.3\}$.
- Smooth initial data: Gaussian pressure pulse (right/left moving acoustic waves + advection).
- Reference: exact solution (analytic in Fourier space) / very fine time step.
- Study:
  1. convergence in h for each scheme (fixed fine grid),
  2. error vs. acoustic Courant number $c\,h/\Delta x$ up to $\gg 1$,
  3. behaviour near resonance $ckh = 2\pi n$,
  4. work–precision (number of evaluations of $\mathcal{L}$, $\mathcal{N}$, or substeps).

Lawson-RK4 on the staggered grid: code

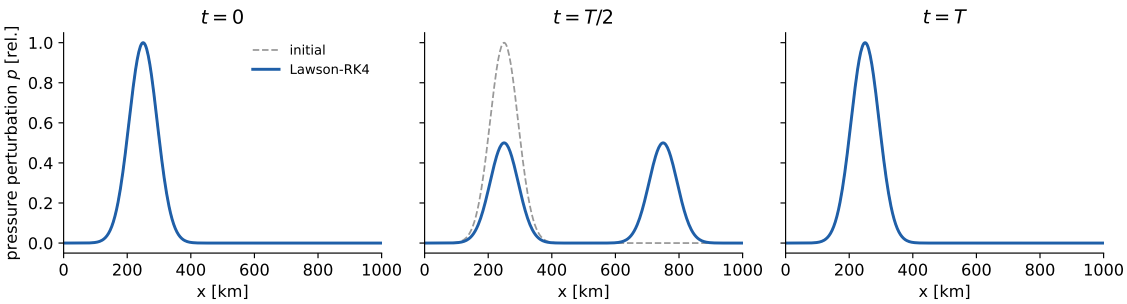
```
def E(p, u): # apply exp(h/2 L), one 2x2 block per wavenumber
    ph, uh = rfft(p), rfft(u) # C, aS, bS: precomputed (cos, sin/omega)
    return irfft(C*ph + aS*uh, nx), irfft(bS*ph + C*uh, nx)

def step(p, u): # 4 evaluations of N, 4 applications of E
    ap, au = N(p), N(u) # k1
    wp, wu = E(p, u) # w = E y
    y2p, y2u = E(p + h/2*ap, u + h/2*au) # Y2
    bp, bu = N(y2p), N(y2u) # k2
    cp, cu = N(wp + h/2*bp), N(wu + h/2*bu) # k3 (Y3 = w + h/2 k2)
    y4p, y4u = E(wp + h*cp, wu + h*cu) # Y4
    dp, du = N(y4p), N(y4u) # k4
    sp = wp + (y2p - wp)/3 + h/3*(bp + cp)
    su = wu + (y2u - wu)/3 + h/3*(bu + cu)
    sp, su = E(sp, su)
    return sp + h/6*dp, su + h/6*du
```

- N: 5th order upwind flux (`np.roll`, real space), E: 4th order staggered acoustic operator, exact exponential.
- Checked against a naive Lawson-RK4 (8 exponentials): identical to round-off.

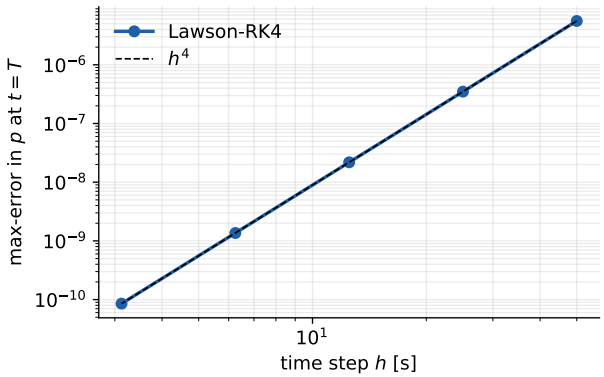
1D experiment: pressure pulse

$c = 350\,\mathrm{m/s}$, $U = 50\,\mathrm{m/s}$ ($M = 1/7$), $L = 1000\,\mathrm{km}$, 400 cells, $T = 10^4\,\mathrm{s}$: $(c+U)T = 4L$ and $(c-U)T = 3L$, so both acoustic pulses return to the start. Initial p: Gaussian, $u = 0$. $h = 50\,\mathrm{s}$: $ch/\Delta x = 7$, $Uh/\Delta x = 1$.



- $t = T/2$: the pulse has split into two half-amplitude pulses; the right-going one has traveled $2L$, the left-going one $1.5L$.
- $t = T$: back at the initial position; max. deviation $6 \cdot 10^{-6}$ (spatial dissipation of the upwind stencil plus time error).

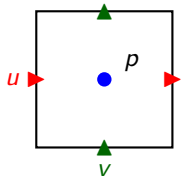
1D experiment: convergence of Lawson-RK4



- Error against the exact semi-discrete solution at $t = T$.
- Observed order 4.00 for $h = 50\,\mathrm{s}$ down to $3\,\mathrm{s}$ (acoustic Courant number $7 \to 0.4$).
- Here $\mathcal{L}$ and $\mathcal{N}$ commute, so the interaction-picture solution is $\mathrm{e}^{t\mathcal{N}}\mathsf{q}_0$ and the error does not grow with c (same max-error for $c = 350$ and $700\,\mathrm{m/s}$).
- Next: variable wind $U(x)$ and 2D, where $[\mathcal{L}, \mathcal{N}] \neq 0$.

Two dimensions

$$\partial_t \begin{pmatrix} p \\ u \\ v \end{pmatrix} = - \begin{pmatrix} \rho_0 c^2 (\partial_x u + \partial_y v) \\ \frac{1}{\rho_0} \partial_x p \\ \frac{1}{\rho_0} \partial_y p \end{pmatrix} - \left(U \partial_x + V \partial_y \right) \begin{pmatrix} p \\ u \\ v \end{pmatrix}.$$



- Arakawa C-grid: p at centers, u on vertical faces, v on horizontal faces.
- Advection of u and v requires *different* stencils in x and y; high-order upwind per direction.
- Fast operator $\mathcal{L}$: 2D wave operator; $e^{h\mathcal{L}}$ via FFT (periodic) or substepping.
- Tests: acoustic pulse in a uniform wind (U, V); rotated wind to test directional splitting errors.

Expected observations (hypotheses to check)

- Split-explicit: simple, but accuracy limited by frozen slow forcing; error scales with the substep count and the acoustic Courant number.
- ETD/Lawson: acoustic stability for arbitrary h, order determined by the outer RK/ETD scheme.
- Resonance steps $ckh \approx 2\pi n$: visible spikes in the error for Lawson/ETD; reduced for averaging; removed for resonance-based schemes.
- When $[\mathcal{L}, \mathcal{N}] \neq 0$ (variable wind, flux-form advection), the commutator gives the leading splitting error; averaging/resonance corrections remove it to higher order.
- In 2D: extra error from the directional splitting of $\mathcal{N}$; cost advantage of FFT-based exponentials shrinks with non-periodic boundaries.

(to be replaced by findings once the experiments are run)

Summary and outlook

- The linear acoustic–advection system isolates the main stiffness/multirate difficulty of NWP dynamical cores.
- **Split-explicit** methods are ETD1-type methods with substepped approximations of $\mathrm{e}^{h\mathcal{L}}$ and $\varphi_1(h\mathcal{L})$.
- **Lawson**, **averaging** and **resonance-based** methods treat the fast flow exactly and differ in how the slow interaction integral is approximated.
- Application: staggered grids in 1D/2D with high-order advection; Lawson-RK4 reaches fourth order, with an error that does not grow with the acoustic Courant number for commuting operators. Variable wind makes the differences between the methods visible.
- Outlook: variable coefficients, orography/terrain-following coordinates, nonlinear (Euler) equations, IMEX/semi-implicit comparison, parallel efficiency.

Thank you for your attention!

References (to be completed) I

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Please check the bibliographic details and add the specific references for averaging and resonance-based methods you intend to follow.